Learning Critical Thinking — A Complete Self-Paced Tutorial

> A structured, practical guide to building critical thinking as a daily skill.
> Based on the **Paul–Elder Framework**, cognitive-bias research, and deliberate-practice methods.

Table of Contents

---

1. What Critical Thinking Is

Critical thinking is the disciplined ability to **analyze information, evaluate evidence, and reach reasoned conclusions** — instead of relying on assumption, opinion, or gut feeling.

The premise that makes it necessary: the mind does **not** naturally grasp truth. Thought is routinely biased by personal agendas, interests, and values; people tend to see what they want to see and twist reality to fit preconceived ideas. Because "trying harder" doesn't fix this, you need an explicit *system* of standards you apply on purpose.

Three skills sit at the center:

- **Analysis** — break a claim or argument into its parts.
- **Evaluation** — judge the quality of those parts against standards.
- **Inference** — draw conclusions that the evidence actually supports.

---

2. The Core Framework (Paul–Elder)

Developed by Dr. Richard Paul and Dr. Linda Elder, this is the dominant academic model. It has **three layers**.

Layer 1 — Elements of Reasoning (the anatomy of any thought)

Every act of reasoning contains these eight parts. Naming them lets you dissect any argument.

Element	Ask yourself
Purpose	What am I trying to accomplish?
Question at issue	What is the central question?
Information	What data/evidence am I using?
Interpretation / inference	How am I drawing conclusions?
Concepts	What theories/definitions am I using?
Assumptions	What am I taking for granted?
Implications / consequences	What follows from this?
Point of view	From what perspective am I looking?

Layer 2 — Intellectual Standards (the quality checks)

Run the parts of your reasoning through these nine tests:

- **Clarity** — *the gateway standard.* If a statement is unclear, you can't judge whether it's accurate or relevant.
- **Accuracy** — Is it actually true? How could I check?
- **Precision** — Is it specific enough? ("Jack is overweight" — by one pound or 500?)
- **Relevance** — Does it actually bear on the question?
- **Depth** — Does it address the real complexities, or is it superficial?
- **Breadth** — Have I considered other viewpoints?
- **Logic** — Do the pieces fit together and follow?
- **Significance** — Am I focusing on what matters most?
- **Fairness** — Am I free of vested interest; am I representing others fairly?

Layer 3 — Intellectual Traits (the character you build)

Apply the standards consistently and you develop: **intellectual humility, courage, empathy, autonomy, integrity, perseverance, confidence in reason, and fair-mindedness.**

The goal: these standards' questions become part of your *inner voice* — guiding your reasoning automatically.

---

3. Know Your Enemy — Biases & Fallacies

Two distinct categories: **biases** are unconscious thinking patterns; **fallacies** are faulty argument structures.

Top cognitive biases to master first

- **Confirmation bias** — seeking only evidence that fits what you already believe. *(The most important one.)*
- **Anchoring** — over-relying on the first number/fact you hear.
- **Availability heuristic** — judging likelihood by how easily examples come to mind.
- **Overconfidence / Dunning–Kruger** — overestimating competence when you know little.
- **Hindsight bias** — "I knew it all along" after the fact.

Top logical fallacies to recognize

- **Strawman** — misrepresenting an argument to attack it. *(Antidote: the **steelman** — improve their argument before you respond.)*
- **Ad hominem** — attacking the person, not the argument.
- **False dilemma** — presenting only two options when more exist.
- **Hasty generalization** — concluding from too little data.
- **Post hoc** — assuming "after this, therefore because of this" (correlation ≠ causation).
- **Appeal to authority** — treating an expert as right outside their actual expertise.
- **Bandwagon** — "everyone believes it, so it's true."
- **Red herring** — distracting with something irrelevant.

---

4. The 8-Week Learning Plan

How to use this plan. Each week is a self-contained module with six parts:

- **Learning objectives** — what you should be able to *do* by the end of the week.
- **What to learn** — the actual teaching content: concepts, definitions, and principles.
- **Key terms** — the vocabulary to internalize.
- **Worked examples** — the ideas applied to real situations.
- **Common mistakes** — the traps learners fall into at this stage.

- **Daily practice & deliverable** — what to do each day, and what to produce.

Spend **one week per module**, ~10–20 minutes a day. The skill comes from *repetition and application*, not from reading ahead. Do not skip Weeks 1–3 — awareness, standards, and bias-detection are the foundation everything else rests on.

Week 1 — See Your Own Thinking (Metacognition & Assumptions)

Learning objectives

- Treat your own thoughts as objects you can examine, not facts you simply have.
- Distinguish a *conclusion* from the *assumptions* it rests on.

What to learn

This week teaches **metacognition** — thinking about your thinking. Most poor reasoning isn't caused by stupidity; it's caused by *invisibility*. We don't see the assumptions underneath our conclusions, so we never question them.

- **A belief is the tip of an iceberg.** Below every "obvious" conclusion sit unstated assumptions. Critical thinking begins by dragging those into the light.
- **Two kinds of assumptions:**
  - *Descriptive* — beliefs about how the world **is** ("People won't read a long email").
  - *Prescriptive / value* — beliefs about how things **should** be ("Managers ought to reply within a day").
- **The assumption test:** For any belief, ask *"What must already be true for this to make sense?"* Each answer is an assumption — and each one is a place the reasoning could be wrong.
- **Fact vs. inference vs. judgment:**
  - *Fact:* "The report is 20 pages." (verifiable)
  - *Inference:* "It's too long for executives." (a conclusion drawn from the fact)
  - *Judgment:* "It's a bad report." (an evaluation)

Confusing these three is the single most common reasoning error — and you'll meet it every week.

Key terms: metacognition · assumption · premise · descriptive vs. prescriptive assumption · inference

Worked examples

> **Belief:** "I shouldn't apply for that promotion — I'm not ready."
> **Assumptions surfaced by asking "what must be true for this to make sense?":**
> - "Ready" means meeting ~100% of the criteria. *(Most people are hired at ~60%.)*
> - My self-assessment is accurate. *(It may be distorted by under-confidence.)*
> - Not applying carries no cost. *(It does — the opportunity is lost.)*
>
> **Lesson:** what felt like a *fact* was three contestable *assumptions* stacked together.

> **Belief:** "We can't raise prices — customers will leave."
> **Hidden assumptions:** all customers are equally price-sensitive; competitors are cheaper; our value hasn't grown since the last price. Each is testable — and possibly false.

Common mistakes

- Listing *reasons* instead of *assumptions* (a reason is stated; an assumption is taken for granted).
- Stopping at the first assumption — keep going; there are usually 3–5.
- Treating an inference ("too long") as a fact ("is too long").

Daily practice & deliverable

Each evening, pick one decision or opinion you formed that day and write the assumptions hiding under it.

Deliverable: a list of **7 beliefs**, each with its 2–4 underlying assumptions.

Week 2 — Demand Clarity, Accuracy & Precision (The Gateway Standards)

Learning objectives

- Convert vague claims into clear, precise, checkable ones.
- Ask the right verification question for any factual claim.

What to learn

You cannot evaluate what you cannot first pin down. These three standards are the *gateway* — until a claim passes them, the other six standards can't even be applied.

- **Clarity is first for a reason.** If a statement is unclear, you literally can't tell whether it's accurate or relevant — you don't yet know what it's saying. The clarity tools: *elaborate* ("Could you say more?"), *illustrate* ("Can you give an example?"), *restate* ("In other words…?").
- **Accuracy = is it true?** A statement can be perfectly clear but false ("Most dogs weigh over 300 pounds"). The accuracy habit is to immediately ask **"How could I check this?"** and identify the source.

- **Precision = is it specific enough to act on?** A claim can be clear *and* true but uselessly vague: "Jack is overweight" — by one pound or fifty? Precision adds the numbers, dates, scope, and thresholds that make a claim operational.
- **The quantify reflex.** Whenever you hear "many," "soon," "better," "most," or "a lot," ask: *how many? by when? better by what measure?* Vague quantifiers hide weak reasoning.
- **Operationalizing.** To "operationalize" a claim is to define it so it can be measured ("productive" -> "tickets closed per week at equal quality"). This is the bridge from opinion to evidence.

Key terms: gateway standard · operationalize · quantifier · verifiability · source vs. claim

Worked examples

> **Vague:** "Our customer satisfaction is dropping."
> **Clear + precise:** "Our post-support CSAT fell from 4.6 to 4.1 (of 5) between Q1 and Q2, concentrated in the enterprise segment."
> **Accuracy check:** Same survey, scale, and population both quarters? Is the sample big enough to be meaningful, or is it noise?

> **Vague:** "The new feature is performing well."
> **Operationalized:** "Adoption hit 32% of active users in three weeks, vs. our 20% target." Now "well" means something — and can be challenged.

Common mistakes

- Accepting a number without asking where it came from (precise ≠ accurate).
- Adding false precision — a made-up exact figure is worse than an honest range.
- Skipping clarity because the claim *feels* obvious.

Daily practice & deliverable

Take one claim a day (headline, email, post) and run it through clarity -> accuracy -> precision.

Deliverable: rewrite **5 vague statements** into clear, precise, checkable ones, each with its verification question.

Week 3 — Hunt Your Confirmation Bias (Evidence Evaluation)

Learning objectives

- Actively seek disconfirming evidence for your own beliefs.
- Weigh evidence by quality, not by whether it agrees with you.

What to learn

Confirmation bias is the highest-leverage target in all of critical thinking: the tendency to seek, notice, and over-value evidence that fits what you already believe — while ignoring or discounting what doesn't. It affects everyone regardless of intelligence, and it quietly corrupts every other step of reasoning.

- **The core counter-move:** ask **"What would change my mind?"** *before* you research. If nothing could, you're not reasoning — you're rationalizing.
- **Seek the strongest counter-evidence, not the weakest.** Disproving a flimsy version of the opposing view ("knocking down the easy ones") leaves your belief untested.
- **A hierarchy of evidence** (strongest -> weakest):
    1. Controlled studies / replicated data
    2. Large representative datasets
    3. Expert consensus (within the actual field of expertise)
    4. Single expert opinion
    5. Anecdote / personal experience
    6. Assertion with no support

Most arguments you'll meet live near the bottom — treat them accordingly.

- **Correlation ≠ causation** (preview of Week 6's *post hoc*). Two things moving together may share a hidden third cause, or be coincidence.
- **Source quality questions:** Who produced this, and what's their incentive? Is it primary (original data) or secondary (someone's summary)? Can it be independently verified?
- **Calibration over certainty.** The goal isn't to flip every belief — it's to hold each one at a confidence level the evidence justifies. "Probably true, under these conditions" is often the honest answer.

Key terms: confirmation bias · disconfirming evidence · hierarchy of evidence · primary vs. secondary source · calibration · base rate

Worked examples

> **Belief:** "Remote work is more productive than office work."
> **Strongest counter-evidence I forced myself to find:** juniors may learn faster in person; some teams show weaker collaboration remotely; *selection bias* — productive people may simply choose remote.
> **Outcome:** belief downgraded from "obvious fact" to "true under specific conditions." That downgrade *is* the skill.

> **Claim seen online:** "City added bike lanes and accidents rose 10% — bike lanes are dangerous."
> **Evaluation:** correlation, not causation. Did cycling volume rise too (more riders -> more incidents even if *per-rider* risk fell)? What's the base rate? The headline cherry-picks one number.

Common mistakes

- "Researching" by typing your belief into a search bar and reading only the agreeing results.
- Treating one vivid anecdote as outweighing a large dataset (availability bias).
- Demanding more proof from claims you dislike than from claims you like (*disconfirmation bias*).

Daily practice & deliverable

Pick one strongly-held belief daily; spend 10 minutes finding the *best* case against it.

Deliverable: a running **bias log** — note each time you catch yourself anchoring, cherry-picking, or generalizing. Naming a bias is the first step to defusing it.

Week 4 — Steelman the Other Side (Intellectual Empathy)

Learning objectives

- Restate an opposing argument so well its holder would say "yes, exactly."
- Separate *understanding* a view from *agreeing* with it.

What to learn

This week builds **intellectual empathy** and **fairness** — the ability to enter a viewpoint you don't share and reconstruct it accurately.

- **Strawman vs. steelman.** A *strawman* distorts an argument into a weak caricature so it's easy to knock down. A *steelman* does the opposite: you build the **strongest, most charitable** version of the other side's case *before* you respond to it.
- **The principle of charity.** Assume the other person is rational and has reasons. Ask: *"What would a thoughtful, informed person who holds this view know that I might not?"*
- **Why it makes* you *stronger:** if you can only beat the weak version of an opposing view, you haven't earned your own position. Steelmanning stress-tests your beliefs and often reveals partial truths on the other side.
- **Ideological Turing test.** The gold standard: can you state the opposing view so convincingly that a neutral observer couldn't tell you actually disagree? If not, you don't yet understand it.
- **Separate the person from the position.** Disagreeing with an argument is not attacking the human making it (this also inoculates you against *ad hominem*, Week 6).

Key terms: steelman · strawman · principle of charity · intellectual empathy · ideological Turing test

Worked examples

> **Situation:** a coworker opposes adopting a new tool.
> **Strawman:** "He just hates change."
> **Steelman:** "Migration has real costs — retraining, data loss, workflow disruption — and the gains are unproven for *our* use case. The risk-adjusted return may be negative in the short term." -> Now you can engage the real argument, and may find he's partly right.

> **Topic:** a policy you personally oppose.
> **Steelman drill:** write three sentences defending it that a sincere supporter would endorse. If you can't, that's a signal you're arguing against a caricature, not the actual position.

Common mistakes

- Sneaking a weakness back in ("Their *best* argument is this dumb thing…") — that's still a strawman.
- Confusing steelmanning with surrendering; you can fully disagree *after* representing it fairly.
- Empathizing with the person but still distorting their logic.

Daily practice & deliverable

For any view you disagree with, write its strongest version before critiquing.

Deliverable: 3 steelmanned arguments for positions you personally reject.

Week 5 — Add Depth & Breadth (Multi-Perspective Analysis)

Learning objectives

- Surface the complexities a surface-level take ignores (depth).
- Analyze an issue from several genuinely different viewpoints (breadth).

What to learn

A conclusion can be clear, accurate, precise, *and* relevant — and still be **shallow** or **one-sided**. Depth and breadth fix those two failures.

- **Depth = engaging complexity.** A slogan like "Just say no" to drugs is clear, accurate, precise, and relevant — yet shallow, because it ignores the real complexities of the problem. Depth asks: *What makes this hard? What second-order effects, trade-offs, and edge cases are being glossed over?*
- **Breadth = multiple legitimate perspectives.** Reasoning can be deep but still trapped in a single viewpoint. Breadth asks: *Who else has a stake? How would this look from their position?*
- **Stakeholder mapping.** List everyone affected by a decision and articulate each one's interests, constraints, and likely objections. Perspectives worth rotating through: the decision-maker, the person who bears the cost, the customer/end-user, the skeptic, and "future you" looking back.

- **Trade-off thinking.** Mature reasoning replaces "Is this good or bad?" with "**Good for whom, at what cost, compared to what alternative?**" Almost nothing is purely good or bad — there are only trade-offs.
- **Second- and third-order effects.** Ask *"And then what?"* repeatedly. The first effect is usually obvious; the consequences that matter often appear two steps downstream.

Key terms: depth vs. surface · breadth · stakeholder analysis · trade-off · second-order effect · steel-thread (following a consequence chain)

Worked examples

> **Issue:** "Should we require a 4-day return-to-office?"
> **Depth (ignored complexities):** commute equity, childcare logistics, role-by-role differences, real-estate cost, measured vs. *perceived* productivity, retention risk.
> **Breadth (3 viewpoints):**
> • *Executive:* culture, collaboration, lease ROI.
> • *Working parent:* schedule rigidity, childcare cost.
> • *New grad:* the in-person mentorship that accelerates early careers.
> A recommendation ignoring any of these is incomplete.

> **Decision:** "Cut the cheapest supplier's contract to save 8%."
> **Second-order "and then what?":** quality dips -> support tickets rise -> churn climbs -> the 8% saving is erased. Depth catches what the spreadsheet misses.

Common mistakes

- Faking breadth by listing viewpoints that all secretly agree with you.
- Mistaking *more words* for *more depth* — depth is new complexity, not elaboration.
- Stopping at first-order effects.

Daily practice & deliverable

Take one issue; (a) list the complexities most people skip, (b) write it from three genuinely different viewpoints.

Deliverable: one issue analyzed for **depth + three-perspective breadth**.

Week 6 — Spot Fallacies in the Wild (Argument Diagnosis)

Learning objectives

- Recognize the most common faulty argument structures on sight.
- Explain *why* each one fails, not just name it.

What to learn

A **logical fallacy** is an error in the *structure* of an argument — the conclusion doesn't actually follow from the premises, even if it sounds persuasive. Learn to spot the pattern and the persuasion loses its grip. Focus on these high-frequency ones:

- **False dilemma** — only two options offered when more exist. *Tell:* "Either X or Y" with no middle ground.
- **Ad hominem** — attacking the arguer instead of the argument. *(Note: a person's credibility can be relevant — but their character doesn't make a* claim *true or false.)*
- **Strawman** — refuting a distorted version of the argument.
- **Hasty generalization** — a sweeping conclusion from too small or unrepresentative a sample.
- **Post hoc ergo propter hoc** — "B happened after A, so A caused B." Confuses sequence with causation.
- **Appeal to authority** — citing an expert outside their field, or treating authority as proof rather than evidence.
- **Bandwagon (appeal to popularity)** — "everyone believes/does it, so it's right."
- **Red herring** — introducing an irrelevant point to distract from the real question.
- **Slippery slope** — claiming one small step inevitably leads to an extreme outcome, with no support for the chain.
- **Circular reasoning** — the conclusion is smuggled into the premise ("It's the best because nothing's better").

Key nuance: spotting a fallacy shows an argument is *poorly supported* — it does **not** prove the conclusion is false. (Believing otherwise is itself a fallacy — the "fallacy fallacy.") A bad argument for a true claim is still a bad argument; look for a *better* one.

Key terms: fallacy · premise/conclusion · validity vs. soundness · correlation vs. causation · fallacy fallacy

Worked examples

> • "Either we cut the budget or the company fails." -> **False dilemma** (raise revenue? reprioritize?).
> • "That study's industry-funded, so it's wrong." -> **Ad hominem / genetic fallacy** (funding warrants scrutiny, not automatic dismissal).
> • "Sales rose after the new logo, so the logo caused it." -> **Post hoc** (ignores season, promotions, market).
> • "Everyone's switching to this framework, so we should too." -> **Bandwagon.**
> • "If we allow remote Fridays, soon nobody will come in at all." -> **Slippery slope** (the inevitable chain is asserted, not shown).

Common mistakes

- Using "that's a fallacy!" as a rhetorical weapon instead of engaging the point.
- Committing the **fallacy fallacy** — assuming a poorly-argued claim must be false.

- Mislabeling a *relevant* credibility point as ad hominem.

Daily practice & deliverable

Find one fallacy a day in real content (ads, op-eds, meetings, social media); name it and explain the flaw.

Deliverable: a log of **7 real-world fallacies**, each named and explained.

Week 7 — Reason From Evidence (Argument Construction)

Learning objectives

- Build your own arguments in a clean, defensible structure.
- State your uncertainty explicitly instead of overclaiming.

What to learn

The first six weeks trained you to *take apart* reasoning. Now you learn to *build* it. A strong argument has four visible parts:

1. **Claim** — a precise, *falsifiable* statement (one that could in principle be shown wrong).
2. **Evidence** — the data or source supporting it, weighted by the Week 3 hierarchy.
3. **Reasoning (the warrant)** — the explicit link explaining *why* the evidence supports the claim. This is the step people skip most — and where arguments quietly break.
4. **Caveat / qualifier** — the conditions under which the claim holds, and what would weaken it.

- **The warrant is the load-bearing wall.** "Sales rose, so the campaign worked" hides a warrant ("nothing else could have raised sales") that's probably false. Always make the connecting logic explicit.
- **Falsifiability.** If no possible evidence could ever count against your claim, it's not an argument — it's an article of faith. Strong claims name their own defeaters.
- **Qualifiers signal strength, not weakness.** "Likely," "in most cases," "given current data" make a claim *more* credible, because they show calibrated confidence rather than bluster.
- **Steel-thread your own case.** Before presenting, attack it yourself: where's the weakest evidence? what's the most likely objection? Address it preemptively.

Key terms: claim · evidence · warrant · qualifier/caveat · falsifiability · defeater · soundness

Worked example

> 1. **Claim:** "We should fix onboarding before adding new features."
> 2. **Evidence:** 40% of trial users drop off in the first session; exit surveys cite confusion.
> 3. **Reasoning (warrant):** features can't help users who never get past setup; fixing the funnel multiplies the value of everything downstream.
> 4. **Caveat:** if drop-off is driven by *price* rather than confusion, onboarding fixes won't move the number — A/B test first.
>
> The caveat is what separates a critical thinker from an advocate.

Common mistakes

- Stating claim + evidence but omitting the *warrant* (the "why it follows").
- Making unfalsifiable claims that no evidence could touch.
- Hiding uncertainty to sound more confident — it makes you *less* trustworthy once a gap shows.

Daily practice & deliverable

Make one claim a day and support it with the four-part structure.

Deliverable: 5 fully-structured arguments, each with an explicit caveat.

Week 8 — Integrate Into Real Decisions (The Full Loop)

Learning objectives

- Run the complete Paul–Elder loop on a decision that actually matters.
- Install a permanent, low-friction habit you'll keep after the program ends.

What to learn

The capstone: combine every prior week into one repeatable decision routine. The aim is *transfer* — moving the skill from worksheet exercises into real judgment under time pressure.

- **The decision walk-through.** For any consequential choice, move through the elements deliberately:
  - **Purpose** — what am I *really* trying to achieve (vs. the surface goal)?
  - **Question at issue** — what exactly am I deciding? (Sharpen it — Week 2.)
  - **Information** — what do I actually *know* vs. *assume*? (Week 1 + 3.)
  - **Standards check** — is my reasoning clear, accurate, relevant, deep, fair? (Weeks 2 & 5.)
  - **Alternatives** — am I in a false dilemma? what other options exist? (Week 6.)
  - **Implications** — what follows from each option, two steps out? (Week 5.)
  - **Point of view** — whose perspective am I missing? (Weeks 4 & 5.)

- **Pre-mortem.** Imagine it's a year later and the decision failed. *Why?* Working backward from imagined failure surfaces risks that forward planning hides.
- **The 10-minute rule (the keystone habit).** Before any consequential conclusion, pause and ask three questions:
  1. *What would change my mind?*
  2. *What am I assuming?*
  3. *Whose perspective am I missing?*
- **Decision journaling.** Write down what you decided, why, and your confidence level. Reviewing past entries calibrates your judgment over time and defeats hindsight bias ("I knew it all along").

Key terms: transfer · pre-mortem · decision journal · keystone habit · calibration

Worked example (compressed)

> **Decision:** accept a new role on a different team.
> • *Purpose:* genuine growth — or just escaping a current frustration?
> • *Information:* I know the title and pay; I'm *assuming* the manager is good -> verify by talking to current reports.
> • *Fairness/breadth:* am I idealizing the new role while only seeing flaws in the current one? (A known bias.)
> • *Pre-mortem:* "A year from now this was a mistake because…" -> the work turned out to be the same, just relabeled.
> • *Implications:* short-term learning curve vs. long-term trajectory.
> **Outcome:** a calibrated decision with its risks named in advance — and logged for later review.

Common mistakes

- Running the full loop only on trivial choices and reverting to gut on the big ones.
- Skipping the pre-mortem because the plan "obviously" works.
- Never reviewing past decisions, so judgment never calibrates.

Daily practice & deliverable

Apply the loop (or at least the 10-minute rule) to one real decision a day.

Deliverable: one significant decision **fully worked through the framework**, with a pre-mortem and a confidence rating — the start of an ongoing decision journal.

At-a-Glance: The 8-Week Arc

Week	Theme	Core skill installed
1	Metacognition & assumptions	Seeing your own thinking

Week	Theme	Core skill installed
2	Clarity · accuracy · precision	Pinning claims down
3	Confirmation bias & evidence	Weighing proof honestly
4	Steelmanning	Representing opponents fairly
5	Depth & breadth	Multi-perspective analysis
6	Fallacies	Diagnosing broken arguments
7	Argument construction	Building defensible claims
8	Integration	Deciding under the full framework

---

5. Everyday Habits That Compound

1. **Ask "How do I know this?"** — for every claim, including your own. Trace it to its source.
2. **Separate observation from interpretation.** "She didn't reply" (observation) vs. "She's ignoring me" (interpretation). Most errors live in that jump.
3. **Argue the other side** periodically, out loud or in writing.
4. **Slow down on emotional topics** — strong feeling signals that bias is active.
5. **Read across viewpoints,** not just within your own echo chamber.
6. **Welcome being wrong** — every corrected belief is a net gain.

---

6. Self-Assessment Checklist

Use this to judge any conclusion (yours or someone else's):

- [ ] Is the claim **clear** and **precise** enough to evaluate?
- [ ] Is it **accurate** — and how was it verified?
- [ ] Is the supporting information **relevant** to the actual question?
- [ ] Does it address the real **complexities** (depth)?
- [ ] Have **other viewpoints** been considered (breadth)?
- [ ] Do the conclusions **logically follow** from the evidence?
- [ ] What **assumptions** is it resting on?
- [ ] Am I being **fair**, or is a bias/vested interest steering me?
- [ ] What evidence **would change my mind**?

---

7. Resources

- **Foundation for Critical Thinking** (criticalthinking.org) — authoritative source for the Paul–Elder model; free guides on the standards and elements.
- **"Logical and Critical Thinking"** — free open course, University of Auckland (fallacies & argument analysis).
- **Books:**
  - *Thinking, Fast and Slow* — Daniel Kahneman (cognitive biases)
  - *Asking the Right Questions* — Browne & Keeley (practical argument analysis)
  - *The Demon-Haunted World* — Carl Sagan (the "baloney detection kit")
  - *Calling Bullshit* — Bergstrom & West (spotting misleading data/claims)

Tip: Don't rush. One week per stage, with the daily 10–20 minutes, beats cramming. The skill comes from repetition, not from reading.